

Progress Report

Day 23 — Team Activity Feed

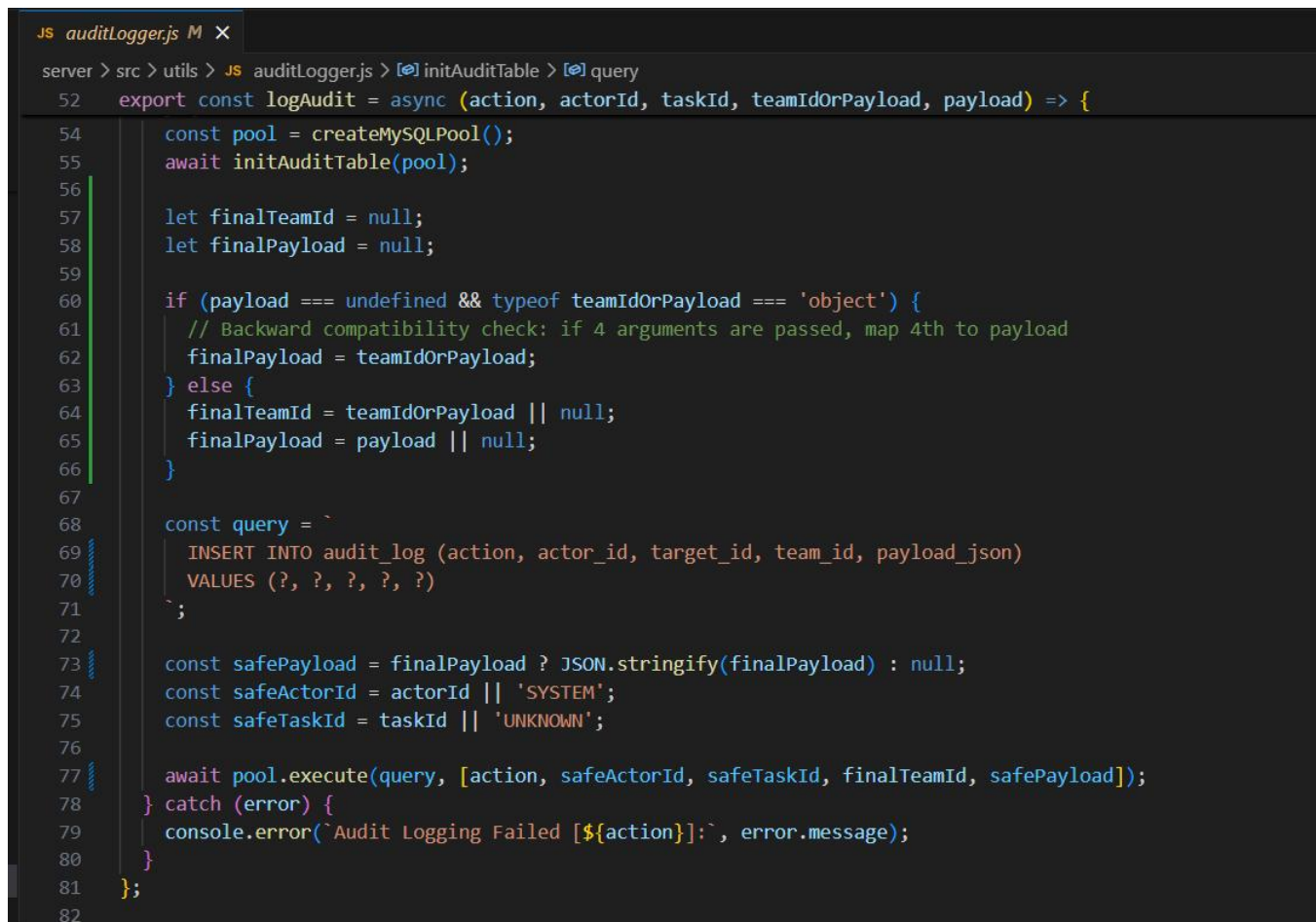
Task Management MERN Stack Platform

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Tasks Completed

Audit Log Enhancements

- Extended the audit logging system to support team-specific activity tracking.
- Added team identifiers to audit records.
- Updated audit logging during task creation, updates, and deletion.

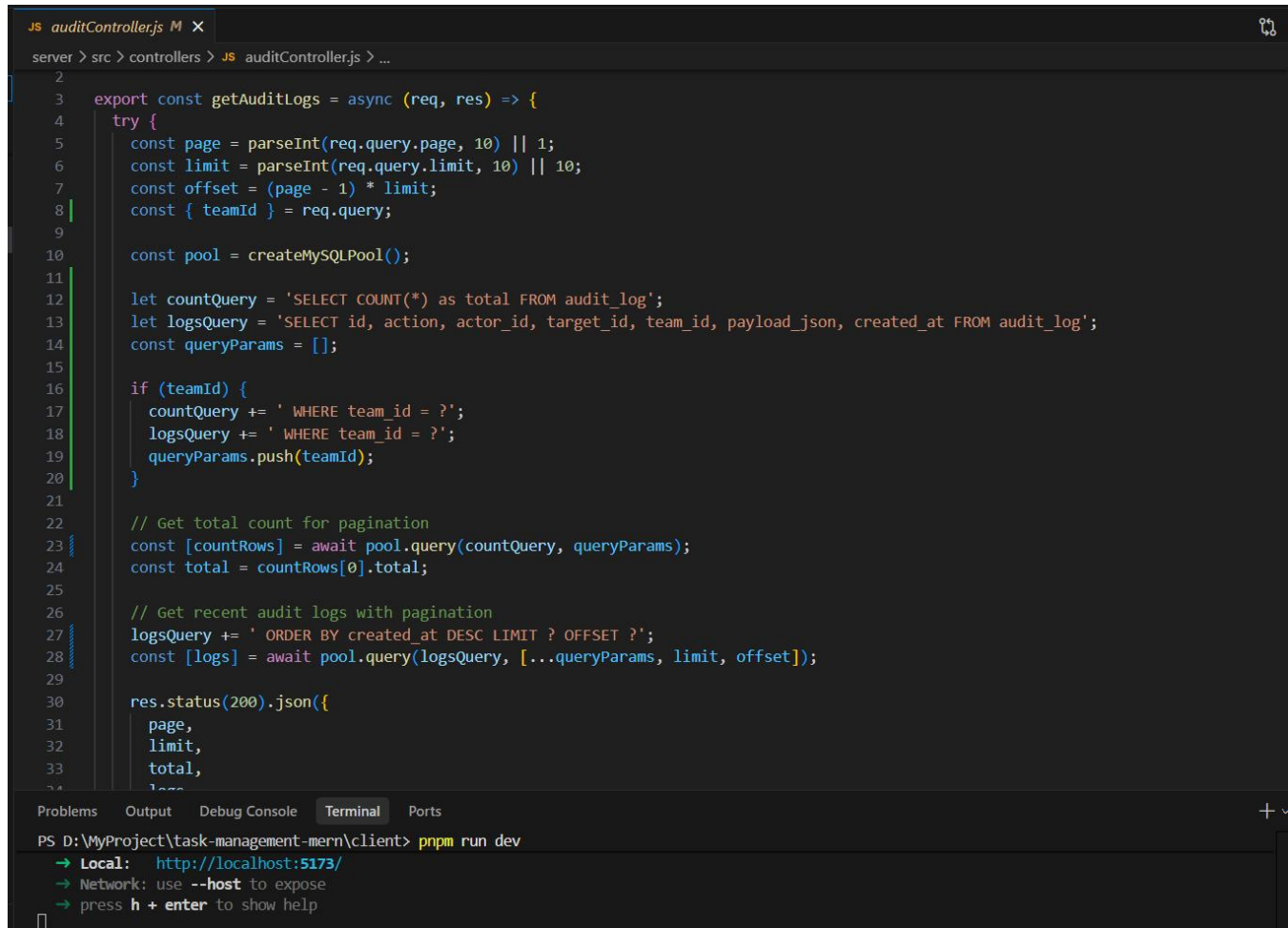


```
JS auditLogger.js M X
server > src > utils > JS auditLogger.js > [0] initAuditTable > [0] query
52 export const logAudit = async (action, actorId, taskId, teamIdOrPayload, payload) => {
53
54   const pool = createMySQLPool();
55   await initAuditTable(pool);
56
57   let finalTeamId = null;
58   let finalPayload = null;
59
60   if (payload === undefined && typeof teamIdOrPayload === 'object') {
61     // Backward compatibility check: if 4 arguments are passed, map 4th to payload
62     finalPayload = teamIdOrPayload;
63   } else {
64     finalTeamId = teamIdOrPayload || null;
65     finalPayload = payload || null;
66   }
67
68   const query = `
69     INSERT INTO audit_log (action, actor_id, target_id, team_id, payload_json)
70     VALUES (?, ?, ?, ?, ?)
71   `;
72
73   const safePayload = finalPayload ? JSON.stringify(finalPayload) : null;
74   const safeActorId = actorId || 'SYSTEM';
75   const safeTaskId = taskId || 'UNKNOWN';
76
77   await pool.execute(query, [action, safeActorId, safeTaskId, finalTeamId, safePayload]);
78 } catch (error) {
79   console.error(`Audit Logging Failed [${action}]:`, error.message);
80 }
81 };
82
```

Figure 2.1 — Team-aware audit logging implemented within the backend audit system.

Team-Specific Audit API

- Enhanced the audit API to support filtering by team.
- Added support for paginated audit log retrieval.
- Configured audit records to be returned in descending chronological order.



```
JS auditController.js M X
server > src > controllers > JS auditController.js > ...

2
3 export const getAuditLogs = async (req, res) => {
4   try {
5     const page = parseInt(req.query.page, 10) || 1;
6     const limit = parseInt(req.query.limit, 10) || 10;
7     const offset = (page - 1) * limit;
8     const { teamId } = req.query;
9
10    const pool = createMySQLPool();
11
12    let countQuery = 'SELECT COUNT(*) as total FROM audit_log';
13    let logsQuery = 'SELECT id, action, actor_id, target_id, team_id, payload_json, created_at FROM audit_log';
14    const queryParams = [];
15
16    if (teamId) {
17      countQuery += ' WHERE team_id = ?';
18      logsQuery += ' WHERE team_id = ?';
19      queryParams.push(teamId);
20    }
21
22    // Get total count for pagination
23    const [countRows] = await pool.query(countQuery, queryParams);
24    const total = countRows[0].total;
25
26    // Get recent audit logs with pagination
27    logsQuery += ' ORDER BY created_at DESC LIMIT ? OFFSET ?';
28    const [logs] = await pool.query(logsQuery, [...queryParams, limit, offset]);
29
30    res.status(200).json({
31      page,
32      limit,
33      total,
34      logs
35    });
36  } catch (error) {
37    res.status(500).json({ error: error.message });
38  }
39 }
```

Problems Output Debug Console Terminal Ports

PS D:\MyProject\task-management-mern\client> npm run dev

→ Local: http://localhost:5173/

→ Network: use --host to expose

→ press h + enter to show help

Figure 2.2 — Team-filtered audit endpoint supporting paginated activity retrieval.

Activity Feed Component

- Developed a reusable Activity Feed component.
- Displayed the latest audit events associated with the selected team.
- Integrated audit data into the Team Detail page.

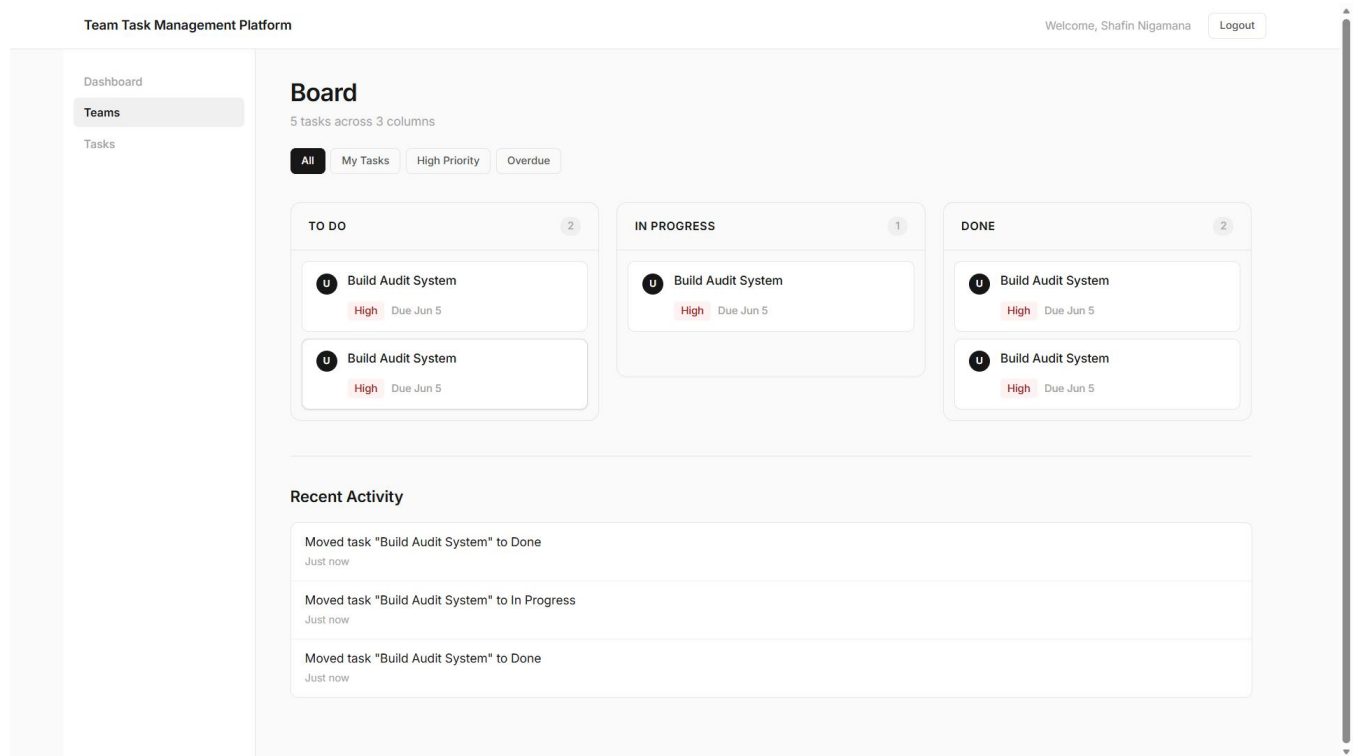


Figure 2.3 — Activity Feed displaying recent team actions.

Activity Event Rendering

- Added readable activity descriptions for task-related events.

Supported events:

- Task Created
 - Task Updated
 - Task Deleted
- Converted raw audit records into user-friendly activity entries.

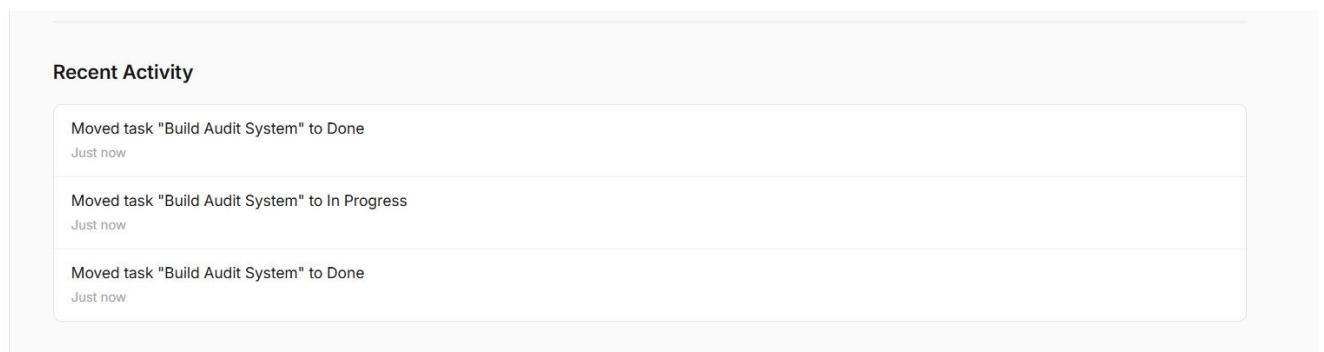


Figure 2.4 — Activity events rendered using human-readable descriptions.

Pagination Support

- Implemented pagination controls for audit history.
- Configured the feed to display the latest 20 records per page.
- Added page navigation for older audit entries.

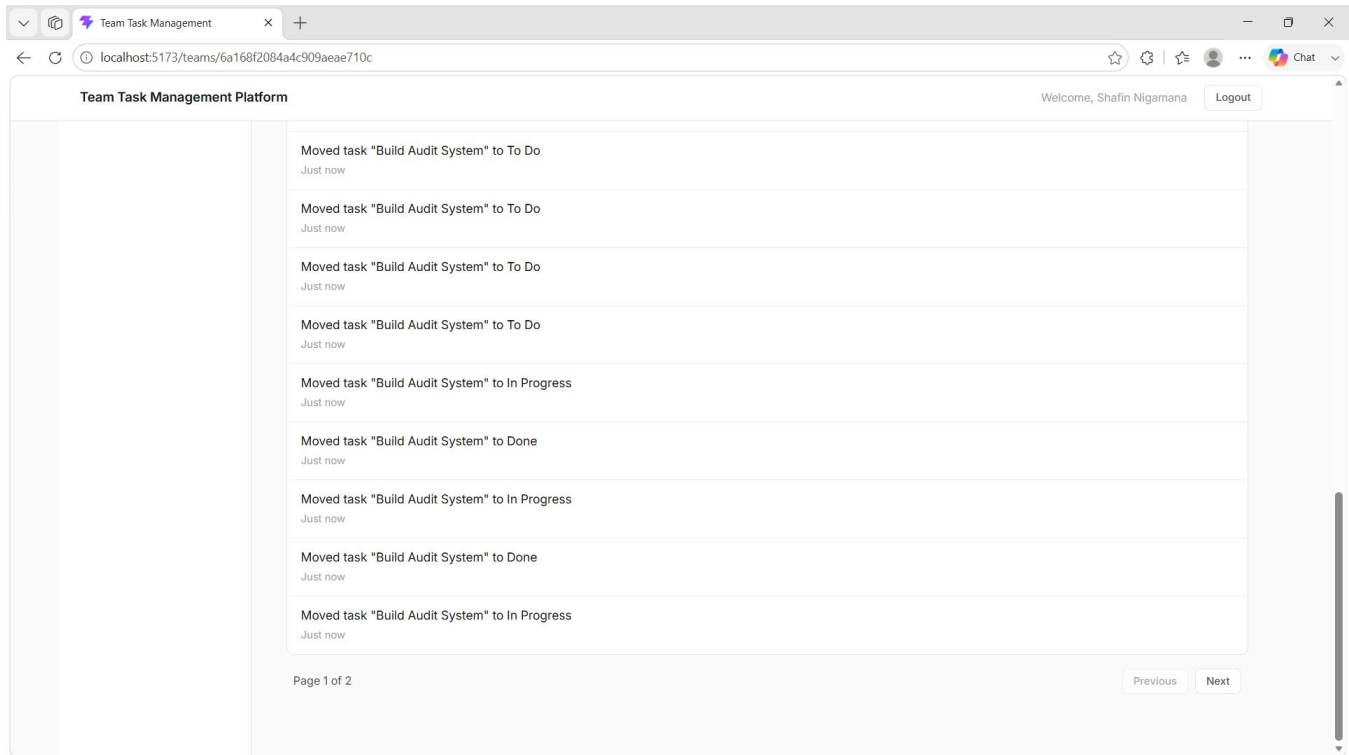


Figure 2.5 — Activity Feed pagination controls for navigating audit history.